

01 / NAMA TIM

TIM CODINGBANG

UI Hackathon 2026 · Challenge Paragon: AI untuk Riset & Prediksi Formulasi · Rimula Agents

ANGGOTA TIM

Nicholas Edmund

TEAM LEAD · AI SAFETY & OPS

Guardian rules (BPOM limits, halal-source flags), Scout evidence from Europe PMC and Open Beauty Facts, VPS release and demo operations, pitch deck.

Marshal Aufa

MACHINE LEARNING & BUSINESS

Designer agent: multi-output Gaussian Process with Bayesian optimization, Virtual Lab simulator, evaluation benchmarks, business case.

Tubagus Dafa

FRONTEND & PRODUCT UX

Formulator dashboard (React + Vite): candidate cards with 95% intervals, live agent timeline, evidence and evaluation tabs.

David Alexander

BACKEND & LOCAL LLM

FastAPI backend and data contracts, Qwen3-30B orchestrator with tool calling and anti-hallucination validation, llama.cpp on GPU.

MEET RIMULA AGENTS

AI-POWERED MULTI-AGENT CO-PILOT FOR SKINCARE R&D

Rimula Agents transforms formulation from a repetitive trial-and-error process into a closed-loop experimentation system that designs, tests, learns, and proposes the next formulation candidate.

BEFORE



AFTER



Rimula
Agents

FROM A PRODUCT BRIEF TO A TESTABLE FORMULA.

Skincare formulation is still a trial-and-error game. Formulators don't need another chatbot; they need a system that turns targets into the next experiment.

SIAPA PENGGUNANYA?



= SKINCARE R&D FORMULATOR

Needs to balance:

PRODUCT TARGET

INGREDIENTS

PROCESS

PERFORMANCE

COST

CONTOH USE CASE

LIGHTWEIGHT BRIGHTENING GEL-CREAM

Target	Requirement
Active	Niacinamide $\geq 4\%$
Viscosity	4,000–12,000 cP
pH	5.0–6.5
Droplet	$\leq 3 \mu\text{m}$
Cost	$\leq \text{Rp}50,000/\text{kg}$
Stability	stable emulsion

MASALAH SEBENARNYA

MULTIPLE TARGETS

Formula harus memenuhi banyak target sekaligus:

pH Viscosity Droplet Stability Active Cost

LARGE SEARCH SPACE

Banyak kombinasi:

Viscosity x Concentrations x Process Parameters

UNCERTAIN OUTCOMES

Formula yang terlihat menjanjikan belum tentu memberikan hasil eksperimen sesuai prediksi.



THE REAL QUESTION IS...

After this experiment, what should we try **next**?

Bukan..

"Can AI generate a formula?"

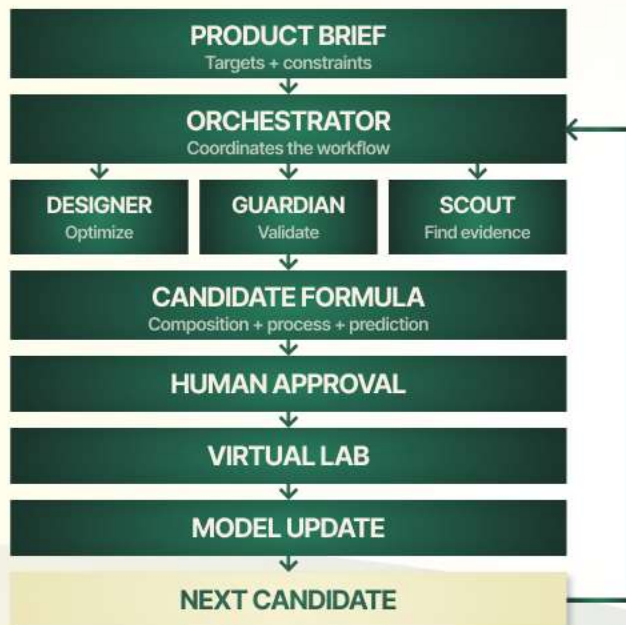
Tetapi..

"Can AI help us decide the next experiment?"

AN AI THAT DOESN'T JUST GENERATE. IT DECIDES WHAT TO TRY NEXT.

Rimula Agents turns formulation into a closed-loop optimization process using specialized AI agents, uncertainty-aware prediction, and experiment feedback.

CLOSED LOOP



3 AI AGENTS

01 · DESIGNER

Find promising candidates

Multi-output Gaussian Process prediction and Bayesian optimization identify candidate compositions and process parameters.

Outputs:

- Formula composition
- Process parameters
- Predicted properties
- 95% uncertainty intervals



02 · GUARDIAN

Check before experimenting

Checks:

- Formulation constraints
- Budget
- pH and active compatibility
- Model uncertainty
- Extrapolation risk
- Regulatory limits and halal-source flags (team assumptions)



PASS

REVIEW

BLOCKED

03 · SCOUT

Bring evidence into the loop

Combines:

- Scientific literature from Europe PMC
- Ingredient and product context from Open Beauty Facts
- Evidence attached to each eligible candidate



MEASURED EVALUATION

Hasil evaluasi terukur

62–78%

fewer batches than random,
across 3 simulator variants

88%

fewer batches in
historical replay

0.93

stability AUC

0.99

viscosity R^2

94–95%

coverage of
95% intervals

306

real liposome experiments:
AI 2.5 vs 6 random steps

Virtual Lab evaluation uses simulation. Liposome validation uses real experimental data but is non-skincare.

Sources: Open Beauty Facts (ODbL) · Zenodo 17867478 liposome data (CC BY 4.0) · Europe PMC · Qwen3-30B-A3B (Apache-2.0) · llama.cpp (MIT)

AI proposes. Human decides. The experiment updates the model.

04 / MOCKUP & DEMO

SEE RIMULA AGENTS IN ACTION

From brief to candidate to experiment to the next model-informed candidate — live on Lintasarta Cloudeka (Deka Notebook, NVIDIA L40S).

01 · DEFINE THE BRIEF

Virtual Lab **Data Historis**

Designer mengusulkan formula baru; formula yang disetujui diuji di Virtual Lab (simulasi), lalu model diarahkan ulang.

BRIEF FORMULASI (TITIK AWAL)

Gel-krim pencerah niacinamide

SPEKIFIKASI TARGET

Target ini bisa diubah sepenuhnya. Brief di atas hanya mengisi angka awal; ubah nilai mana pun sebelum menjalankan.

NIACINAMIDE $\geq$ (%)

4

VIKOSITAS (CP)

4000 12000

PH

5 6.5

DROPLET D50 $\leq$ (MM)

3

BIAYA $\leq$ (RP/KG)

50000

☒ Wajib stabil

JUMLAH KANDIDAT

3 kandidat

☒ Orchestrator LLM (Qwen3 lokal)

Jalankan

Loop belajar (global) **Reset loop**

0 batch diuji 0/6 spesifikasi terbaik belum target tercapai

02 · REVIEW CANDIDATES

Rekomendasi: Lakukan uji laboratorium formulasi untuk stabilitas, viskositas, pH, dan ukuran droplet pada #1 terlebih dahulu, mengingat kandidat ini memiliki dukungan literatur terkuat untuk efikasi pencerahan. Jika hasil uji sesuai, lanjutkan ke #2 dan #3 untuk evaluasi lebih lanjut.

Bandingkan kandidat **Kartu** **Tabel**

KANDIDAT #1 **Perlu ditinjau**

95%

peluang seluruh spesifikasi

Optimal - Biaya Terendah - Rp 29.134

visk **●**

pH **●**

D50 **●**

Peluang stabil 95%

Niacinamide 4% ($\geq$ 4%)

Terbuka di bawah $\downarrow$

KANDIDAT #2 **Perlu ditinjau**

95%

peluang seluruh spesifikasi

Variasi #2 - Rp 29.258

visk **●**

pH **●**

D50 **●**

Peluang stabil 95%

Niacinamide 4% ($\geq$ 4%)

Buka detail $\rightarrow$

KANDIDAT #3 **Perlu ditinjau**

95%

peluang seluruh spesifikasi

Variasi #3 - Rp 30.505

visk **●**

pH **●**

D50 **●**

Peluang stabil 95%

Niacinamide 4,28% ($\geq$ 4%)

Buka detail $\rightarrow$

03 · APPROVE, TEST, LEARN

KEPUTUSAN FORMULATOR

☒ Disetujui : "Peluang spesifikasi tertinggi dan biaya terendah; sertifikat halal pemasok akan diverifikasi."

Hasil Virtual Lab Emulsi - Batch ke-1 **6/6 Spesifikasi Terpenuhi — TARGET TERCAPAI**

PARAMETER	TARGET	PREDIKSI	HASIL
Niacinamide	$\geq$ 4%	4%	4% ✓
Viskositas	4.000-12.000 cP	6.881	8.092 ✓
pH	5-6,5	5,62	5,74 ✓
Droplet D50	$\leq$ 3 μ m	1,49	1,52 ✓
Stabil	stabil	peluang 95%	stabil ✓
Biaya bahan	$\leq$ Rp 50.000/kg	Rp 29.134	Rp 29.134 ✓

Model diperbarui ke **gp-global-v10**

The formulator sets product targets, constraints, candidate count, and whether to use the local orchestrator.

Each candidate includes its formula, predicted properties, uncertainty, Guardian status, and scientific or market evidence.

After human approval, the candidate is tested in the Virtual Lab. The result is recorded as a batch and the model is updated.

CLOSED-LOOP RESULT

Median batches to reach the target — full 20-seed Virtual Lab evaluation.

AI **2.0 batches**

RANDOM BASELINE

9.0 batches

Historical replay: 2.5 batches with AI vs 20 random. Estimated saving per formula (simulation): $\approx$ 7 batches, $\approx$ Rp 10.5 jt, $\approx$ 98 days.

Virtual Lab results are simulation results. Final formulation decisions remain with the formulator and require physical laboratory validation. Pilot at Paragon: replace the Virtual Lab with lab records (same data schema), fully on-premise.

Local Qwen3 orchestration · Gaussian Process prediction · deterministic guardrails · human approval